

Are FIFA Rankings *Actually Predictive?*

Assignment 1

GROUP 1

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Rankings decide things, but nobody checks them

Rankings drive tournament seeding, World Cup qualification slots, and billions in betting markets. But they're rarely tested against what actually happens.

Federations

Use rankings for seeding and qualification slots.

Bookmakers

Price odds based on the ranking signal.

Researchers

Baseline for sports analytics work.

Two public datasets joined on country and date

International Football Results

Kaggle · martj42

49,287 matches

date, home_team, away_team, scores, tournament

FIFA World Ranking

Kaggle · cashncarry

67,472 monthly snapshots

rank, country, total_points, confederation, rank_date

OUR FACT

One match between two national teams on a specific date in a specific tournament, with both teams' FIFA ranking at the time.

HOW THEY CONNECT

Join on country and date. Each match gets the most recent ranking that was available before kickoff.

Four questions, four dimension combos

01

Tournament Type × Date

How well do rankings predict outcomes across tournament types?

This tells analysts, fans, federations, or media whether FIFA rankings are reliable predictors, and whether pressure/tournament intensity makes outcomes less predictable.

02

Year × Match Context

Has home advantage changed over time?

Tests if 'globalization killed home advantage' is real.

03

Confederation × Confederation

Which confederations beat or underperform their ranking?

Reveals where the ranking model has blind spots; whether teams from some confederations are underrated or overrated internationally, which is useful for scouting, tournament forecasting, and evaluating ranking fairness.

04

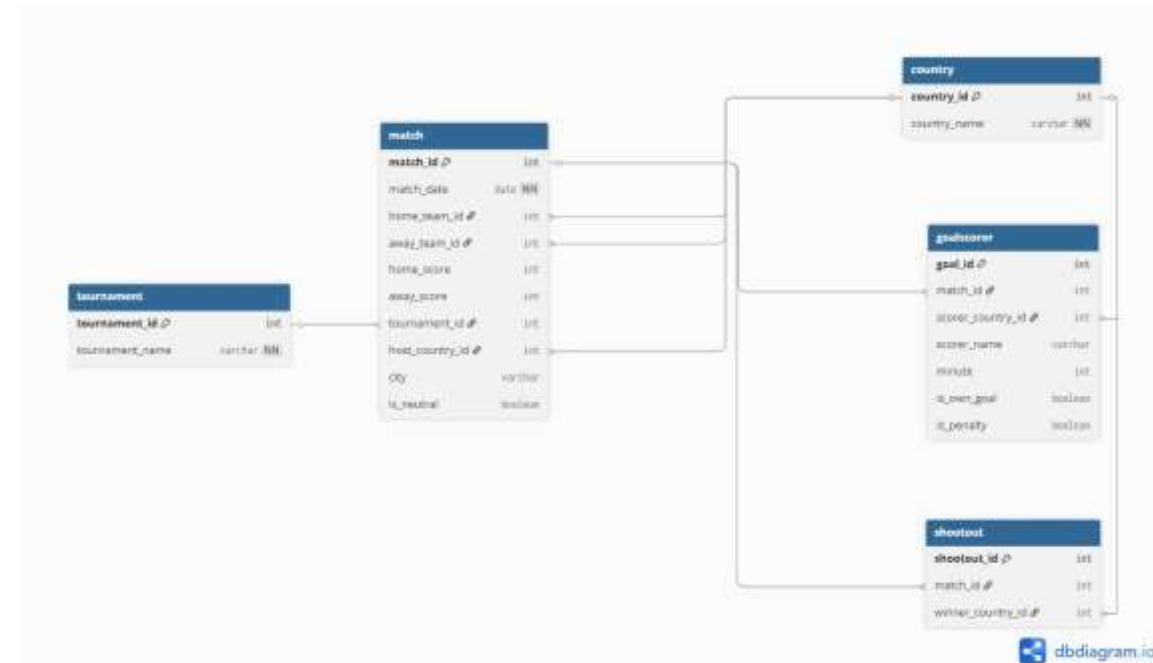
Rank Gap × Top-20 Tier

When are rankings reliable, and when do they fail?

Useful for big gaps but unreliable for close matchups?

Both source datasets in 3NF

Matches dataset



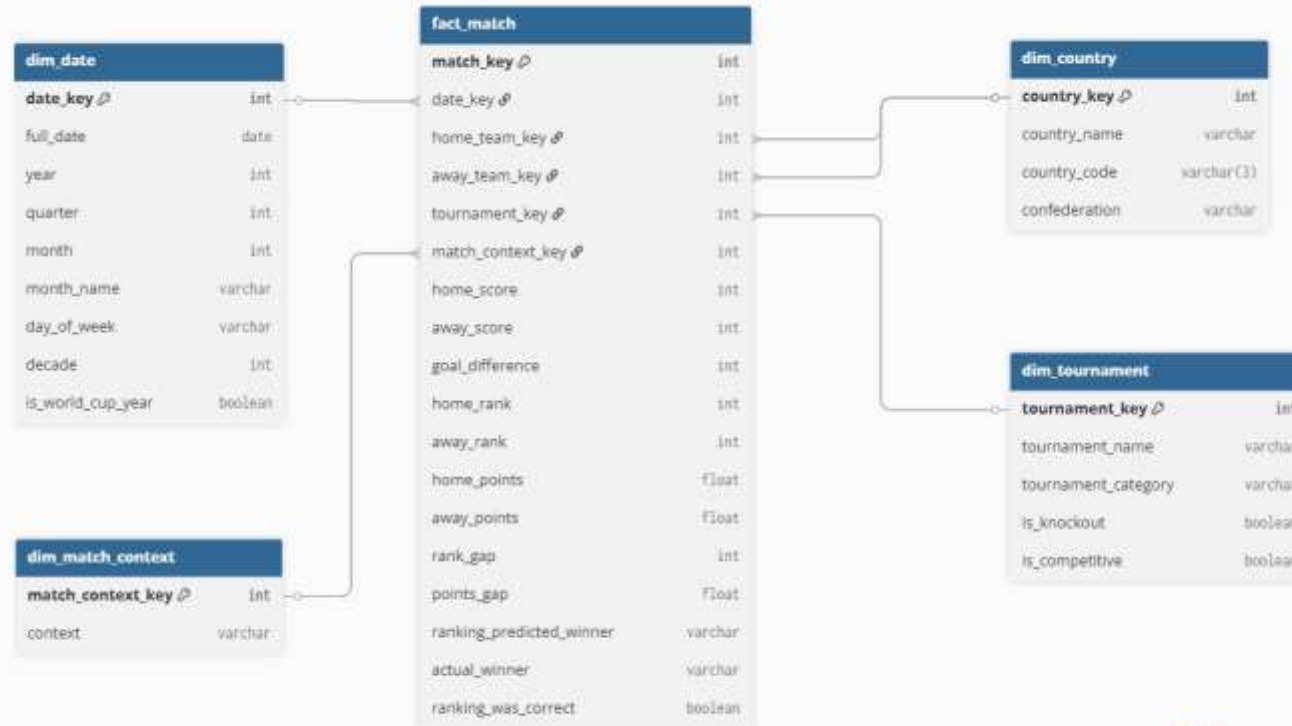
Country and Tournament split out as their own entities.

Rankings dataset



Confederation split out so it doesn't depend on country.

One fact table, four dimensions



DESIGN CHOICES

Grain

One row = one match. With both rankings attached.

Role-playing country

dim_country joins twice: home and away.

Confederation denormalized

Moved into dim_country to keep queries simple.

Python ETL, runs end-to-end with one command



TWO REAL PROBLEMS WE HAD TO SOLVE

Country names didn't match

18 country names were different between the two datasets — Iran vs IR Iran, USA vs United States, Czech Republic vs Czechia. We fixed them with a mapping CSV that documents every rename.

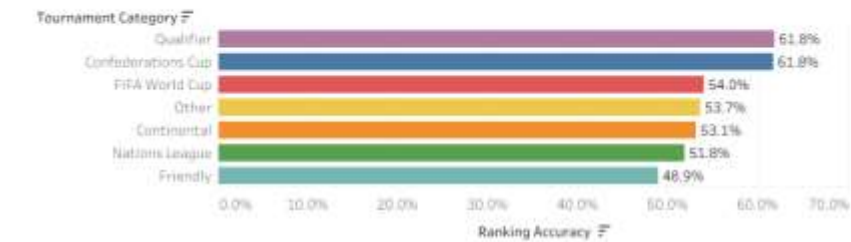
Rankings are monthly, matches are daily

A regular join on equal dates would lose most matches. We used `merge_asof` to grab the most recent ranking before each match — never a future one.

Four charts, one question each

Are FIFA rankings actually predictive?

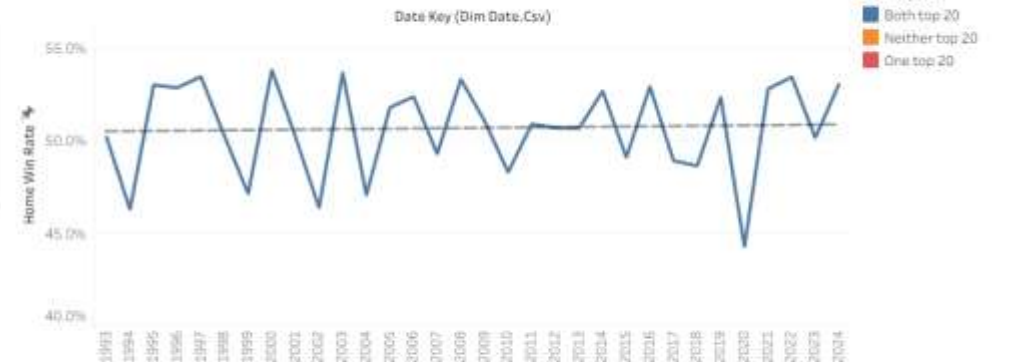
How well do rankings predict outcomes across tournament types?



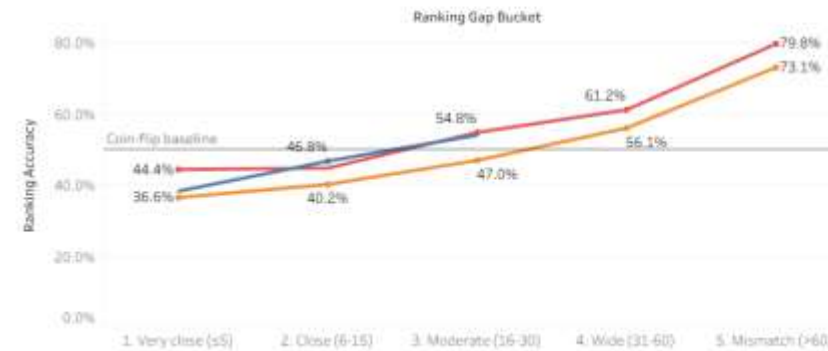
Which confederations beat or underperform their ranking?

Confederati...	AFC	CAF	CONCAC...	CONMEB...	OFC	UEFA
AFC		23.0%	21.9%	16.0%	-0.9%	15.1%
CAF	-6.9%		10.5%	12.3%	13.5%	10.9%
CONCACAF	-5.7%	-3.6%		0.4%	-14.3%	-7.5%
CONMEBOL	-4.4%	-2.7%	-2.1%		-8.3%	-3.9%
OFC	32.8%	40.0%				25.0%
UEFA	7.4%	-1.5%	6.5%	7.3%	-2.4%	

Has home advantage changed over time?



When are rankings reliable, and when do they fail?



LIVE DEMO

Switching to Tableau

Rankings are a coin flip in friendlies

12.9 pp

Gap between qualifiers (61.8%) and friendlies (48.9%)

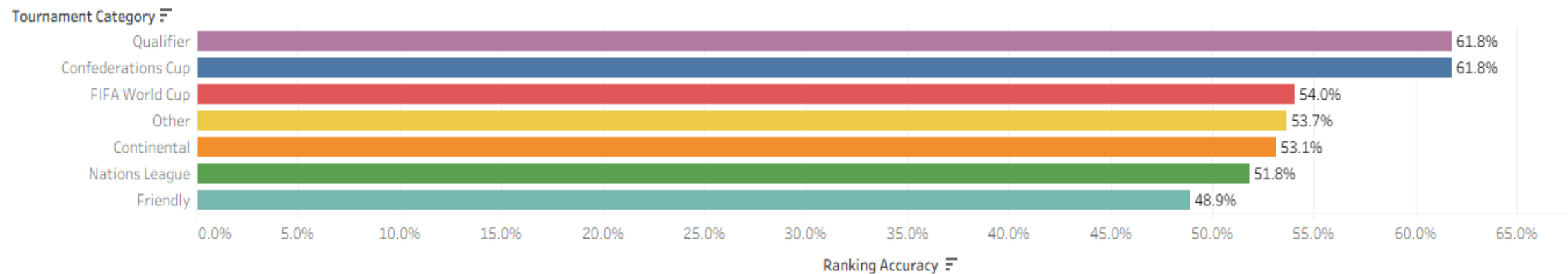
WHAT WE FOUND

Rankings pick the winner correctly 62% of the time in qualifiers, but only 49% in friendlies basically a coin flip. World Cup and Continental tournaments sit in the middle at around 54%.

WHY IT MATTERS

Betting models that trust rankings too much lose money on friendlies. Rankings work when teams play hard, not when they rest starters.

How well do rankings predict outcomes across tournament types?



Home advantage hasn't actually declined

p = 0.81

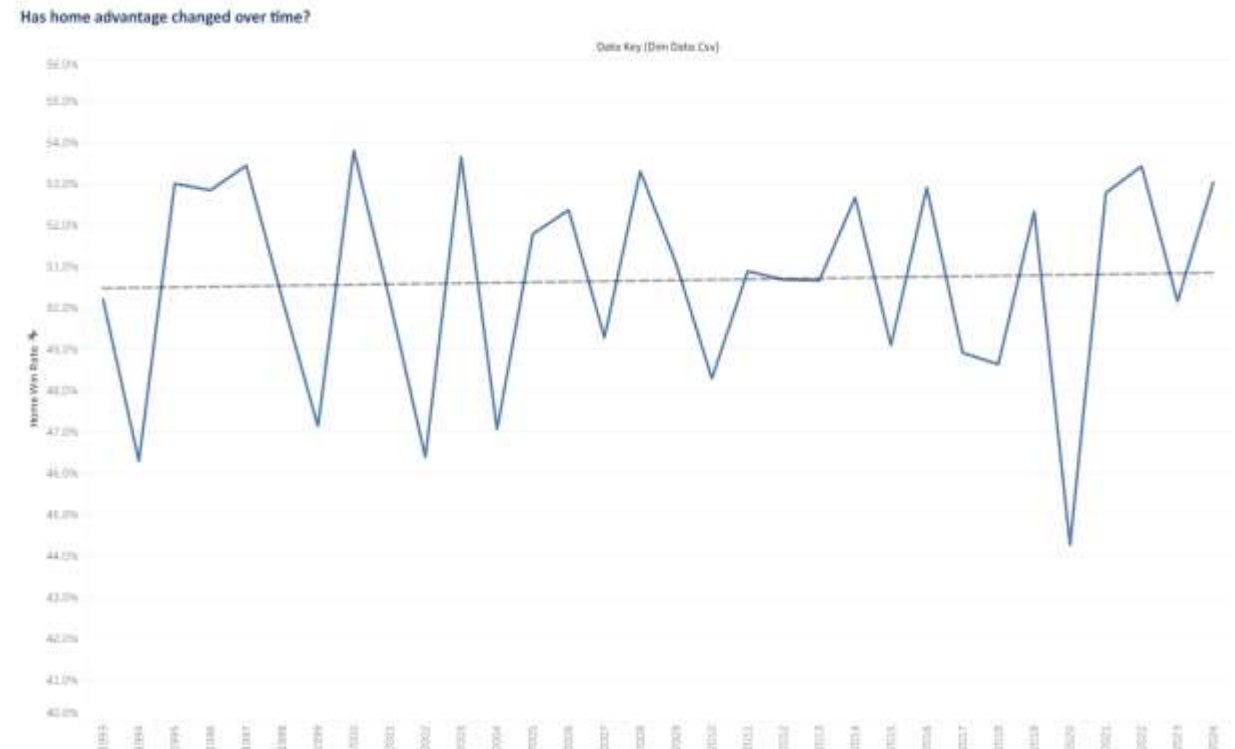
31-year trend is statistically flat ($R^2 = 0.01$)

WHAT WE FOUND

Home teams win around 50–51% of the time across all 31 years. The 2020 dip to 44% was the COVID empty stadium effect, but that recovered within two years.

WHY IT MATTERS

The 'home advantage is dying' isn't in the data. This makes it so that hosts don't need to change fees over an imaginary claim



Rankings underrate African and Asian teams across the board

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Confederations that consistently beat their ranking

WHAT WE FOUND

The AFC and CAF rows are positive against almost every opponent, both confederations beat their FIFA ranking expectations by 10 to 24pp. UEFA and CONMEBOL are rated within a few points of each other. And the matches between OFC and other confederations are mostly noise, there weren't a lot of matches played so the results are a bit misleading.

Which confederations beat or underperform their ranking?

Confederati..	Confederation (Dim Country.Csv1)					
	AFC	CAF	CONCAC..	CONMEB..	OFC	UEFA
AFC		23.8%	21.8%	16.0%	-0.5%	15.1%
CAF	-6.9%		10.5%	12.3%	13.6%	19.9%
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WHY IT MATTERS

Rankings probably overweight regional strength. UEFA teams rack up points against strong neighbors; CAF and AFC teams have fewer strong neighbors. Cross-continental forecasts that ignore confederation will undervalue AFC and CAF teams.

Rankings only predict when the gap is wide

37%

Ranking accuracy for close matchups is below a coin flip

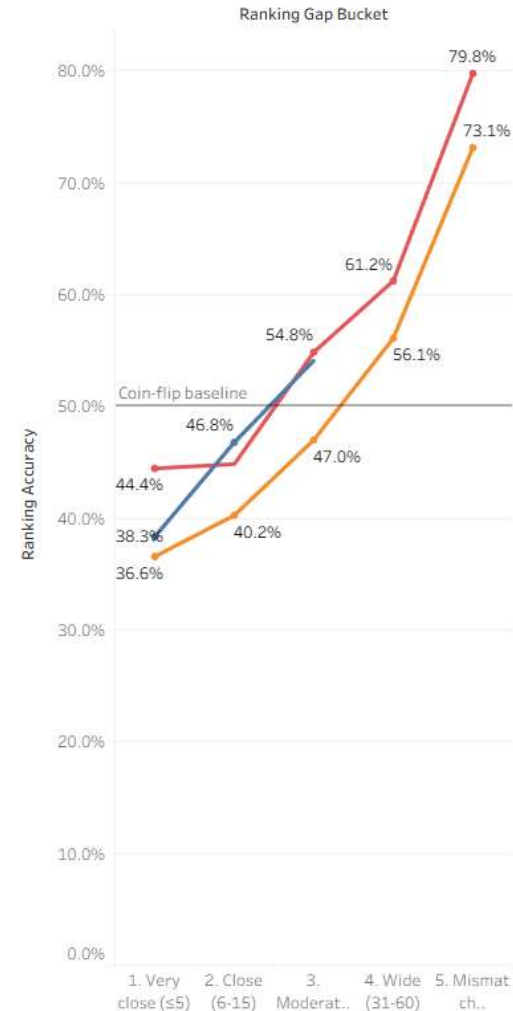
WHAT WE FOUND

For close matchups (gap ≤ 5), rankings only get it right 37–44%, below 50% because draws cluster here. For big mismatches (gap > 60), accuracy jumps to 73–80%. Between two top 20 teams, rankings barely beat chance.

WHY IT MATTERS

Rankings can separate strong teams from weak ones. They can't tell similar teams apart. Tournament seeding between top teams is basically random.

When are rankings reliable, and when do they fail?



What rankings actually do and don't do

THEY WORK WHEN

- The gap is wide (>60): 73 - 80% accuracy
- Matches are important (qualifiers): 62% accuracy, or teams play in major tournaments (World Cup, Continental): 54% accuracy.
- The matchup involves a clearly stronger team: 70%+ accuracy when gap >30

THEY FAIL WHEN

- It's a friendly (49% - basically random)
- AFC or CAF play international matches
- Both teams are top 20

Rankings tell strong teams from weak ones.

They can't tell similar teams apart, and they underrate AFC and CAF.

W R A P - U P

Limitations, repo, questions

LIMITATIONS WE KNOW ABOUT

Pre 1993 and post 2024 matches dropped

Rankings only cover 1992-06 to 2024-06.

Knockout flag is tournament level

Can't split group vs knockout per match. Basically, removed the group stage/quarter-final etc. and just kept the matches on a tournament level

OFC results are noisy

Too few neutral cross confederation matches.

REPOSITORY

[GitHub Repository](#)

Pipeline, raw + clean CSVs, Tableau workbook.

Questions?

Thank you for listening.